

Internet Appendix for

**“Displacement Versus Augmentation: The Effects of AI on
Employment Dynamics and Firm Value”**

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IA.1 Identifying and Classifying AI Patents Using Generative LLMs

IA.1.1 Prompt Engineering

Until recently, the standard approach in finance and economics for analyzing textual data has been to use text filtering or word embedding techniques to classify documents according to single-word frequencies or the co-occurrence of certain pairs of words.¹ While such methods are straightforward and easy to interpret, they do not make full use of the semantic connections between words, sentences, and documents that can dramatically improve textual analysis. The recent development of generative AI tools such as ChatGPT, Claude, and Google Gemini has made it possible to retrieve, categorize, and analyze documents using simple, natural-language queries that refer to general topics or concepts. These instances of large language models (LLMs) are trained on trillions of tokens and can respond to queries using a large set of commonsense knowledge.²

In view of the difficulties with using traditional NLP techniques to capture the rich semantic content in patent filings, we turn to generative AI to help identify and classify different types of AI innovations. Specifically, we use an advanced, open-source LLM called *Qwen2.5-14b-Instruct*. This generative AI model was developed by Alibaba Cloud and released to the public in September 2024 under an open-source license for broad commercial and research use. The model features 14.8 billion parameters and is an instruction-tuned version of a base model that was trained on data consisting of approximately 18 trillion tokens (<https://github.com/QwenLM/Qwen2.5>). According to several benchmark evaluations as of October 2024, *Qwen2.5-14b-Instruct* performs competitively within its size class. In engineering-related tasks, it achieves performance comparable to, and in some cases exceeding, that of larger models such as *Gemini-1.5-Pro*, *Claude 3-Opus*, and *GPT-4 Turbo*.³

Our approach to identifying and classifying AI innovations from among the millions of U.S. patent filings during 2007-2023 consists of two main steps. First, for each patent application, we formulate a query to elicit from the LLM an inference about whether the invention is AI-related.

¹ Some notable examples of this approach to computational linguistics include Tetlock et al. (2008), Loughran and McDonald (2011), Jegadeesh and Wu (2013), Hoberg and Phillips (2016), and Li et al. (2021).

² An emerging literature in finance and economics applies LLMs and generative AI techniques for text classification and content analysis. Eisfeldt and Schubert (2024) provide a review of recent finance research that uses generative AI techniques to analyze and classify textual data.

³ See, e.g., the LLM performance benchmarking leaderboard at <https://huggingface.co/spaces/TIGER-Lab/MMLU-Pro> (accessed October 18, 2024).

To reduce ambiguity and to obtain from the LLM as clear a response as possible, we use prompts that feature a specific YES/NO question combined with the text of a patent abstract. For example, to check whether or not U.S. Patent Application #20210209449 is related to artificial intelligence, we prompt the LLM with the following question/instruction, along with the patent abstract as context: *“Is the following invention directly related to artificial intelligence? Respond with just YES or NO. Abstract: Graphical elements in a user interface (UI) may be detected in robotic process automation (RPA) using convolutional neural networks (CNNs). Such processes may be particularly well-suited for detecting ... [remaining abstract text follows]”*. Note that, in this step, we do not require the LLM to provide an explanation for its yes/no inference as doing so would greatly extend the time required for each inference.⁴ Applying this querying procedure to all 5,323,321 patent applications in the sample yields a classification of each patent application as being either AI-related or not.

Next, for each patent application that is deemed to be AI-related, we use seven separate queries to the LLM to ascertain whether or not the application belongs to one or more of the seven AI types. We again use simple yes/no prompts to ensure speed of inference. For instance, to check whether a patent application is related to perception-based AI, we query the LLM with the following prompt: *“Is the following invention directly related to artificial intelligence that has perceptual ability? Respond with just YES or NO. [patent abstract text follows]”*. The overall collection of binary yes/no LLM responses provides a full classification of AI patent applications in our sample.

IA.1.2 Validation

We employ an “LLM-as-a-judge” approach (Zheng, 2023; Gu, 2025; Li, 2025) in which several frontier AI models serve as judges of the correctness of responses from the *Qwen2.5-14b-Instruct* open-source LLM. The judge models we use include the following models: *Gemini 2.5 Pro* (released June 17, 2025 by Google DeepMind); *Claude Opus 4.1* (released August 5, 2025 by Anthropic); *GPT 4.1* (released April 14, 2025 by OpenAI); and *DeepSeek R1 0528* (released May

⁴ As has been well-documented, token generation accounts for a large majority of the processing time required for an LLM to respond to a query. Our approach of eliciting simple yes/no responses from an LLM without requiring explanations has the advantage of dramatically shortening the overall time required for inference. Moreover, we have verified that directing the LLM to forego providing explanations does not seem to compromise the quality of its inferences.

28, 2025 by DeepSeek). As of early August 2025, all four models rank at or near the top of the LMArena leaderboard (<https://lmarena.ai/leaderboard>) from among 234 ranked models in terms of text-based generative AI performance.

We start by assessing the quality of basic inferences made by *Qwen2.5-14b-Instruct* about whether patent applications are AI or non-AI. For each year in the 2007-2023 sample period, we draw a random sample of 500 patent applications, resulting in a total sample of 8,500 applications. For each sampled application, we prompt *Qwen2.5-14b-Instruct* with “*Is the following invention directly related to artificial intelligence? Respond with YES or NO and give an explanation. [patent abstract text follows]*”. We then feed each response given by *Qwen2.5-14b-Instruct* to each of the four judge models using the following prompt: “*You will be presented with a question about an invention and a response to the question by an LLM. Please just state whether the response by the LLM is CORRECT or INCORRECT. [The prompt given to Qwen2.5-14b-Instruct and the resulting response text follow]*”.

For inferences about the seven individual AI types, we assess inference quality by drawing yearly random samples of 500 patent applications (2007-2023) from among those that were previously labeled by *Qwen2.5-14b-Instruct* as being AI-related. Then, for each application and each AI type ($8,500 \times 7 = 59,500$ inferences in total), we prompt *Qwen2.5-14b-Instruct* for a yes/no answer and an explanation of why the patent application does or does not belong to the type in question. Each response from *Qwen2.5-14b-Instruct* is then provided to each of the four judge models in conjunction with a prompt identical to the one used for the basic AI/non-AI categorizations.

From the assessments given by the four judge models, we calculate the accuracy of *Qwen2.5-14b-Instruct*’s overall AI/non-AI inferences as well as the accuracy of its inferences about each of the seven AI types. As seen in Table IA.1, the percentage of AI/non-AI inferences judged to be correct ranges from 96.0% to 98.2%, depending on which judge model is being applied. For inferences about whether patent applications belong to each of the seven AI types, the percentage of inferences judged to be correct ranges from a low of 77.2% (engagement-based AI) to a high of 98.8% (creativity-based AI).

To further validate the reliability of our LLM-based approach, we manually review numerous instances of yes/no classifications and explanations from *Qwen2.5-14b-Instruct*. Appendix IA.2 provides some illustrative examples for each AI type. Our review of various examples suggests

that Qwen2.5-14b-Instruct is indeed capable of differentiating between fine shades of meaning in patent abstracts. For instance, consider the case of Patent Application #20230294736, “Offline Tracking System for Autonomous Vehicle Control Systems,” (GM Cruise Holdings LLC), listed in Appendix IA.2. Despite the fact that the abstract for this patent filing does not contain the term “perception” or any other synonymous terms, the model correctly recognizes that the invention demonstrates perceptual AI ability because the disclosed system can track, identify, and annotate objects in an autonomous vehicle’s environment. Also, we note that the LLM is able to reliably determine when an AI invention does not relate to a particular category. For example, the LLM correctly infers that Patent Application #20210258756, filed by Accenture Global Solutions Ltd., does not directly pertain to engagement with users (see Appendix IA.2). As explained by the LLM, although this invention involves sophisticated data processing and decision-making, its capabilities relate to generating automated alerts for users rather than learning from interactions and engaging with users through AI-driven conversational techniques.

Table IA.1 Using an “LLM-as-a-Judge” Approach to Evaluate Patent Classifications

This table shows the results of using several leading, commercially-available AI models to judge the correctness of inferences from the open-source *Qwen2.5-14b-Instruct* model (Alibaba Cloud). Reported in the table are the percentages of sampled inferences from *Qwen2.5-14b-Instruct* that are assessed as being correct by the following four judge LLMs: *Gemini 2.5 Pro* (Google DeepMind), *Claude Opus 4.1* (Anthropic), *GPT 4.1* (OpenAI), and *DeepSeek R1 0528* (DeepSeek). The sample of inferences by *Qwen2.5-14b-Instruct* consists of a stratified random sample of 500 patent applications drawn from each year during 2007-2023. For the seven AI subtypes, inferences are based on a stratified random sample of 500 patent applications per year drawn from among those applications deemed by *Qwen2.5-14b-Instruct* to be directly related to AI.

AI Type	Judge Model			
	<i>Gemini 2.5 Pro</i>	<i>Claude Opus 4.1</i>	<i>GPT 4.1</i>	<i>DeepSeek R1 0528</i>
Any	97.7%	96.0%	97.0%	98.2%
Perception	92.2%	82.6%	93.6%	95.7%
Motor control	96.6%	95.9%	97.9%	97.9%
Language	95.6%	92.5%	97.1%	97.0%
Decision-making	89.5%	83.0%	91.8%	89.6%
Learning	89.2%	77.3%	84.0%	87.8%
Engagement	90.5%	77.2%	87.7%	91.6%
Creativity	96.7%	94.0%	97.1%	98.8%

IA.2 Examples of AI Patents Identified from LLM Queries

PERCEPTION

Query: *Is the following invention directly related to artificial intelligence that has perceptual ability? Respond with YES or NO and give an explanation.*

Title: OFFLINE TRACKING SYSTEM FOR AUTONOMOUS VEHICLE CONTROL SYSTEMS

Patent Application Number: 20230294736

Patent Applicant: GM Cruise Holdings LLC

Abstract: Disclosed are systems, apparatuses, methods, and computer-readable media to autonomous driving vehicles and, in particular, for tracking objects in an environment that an autonomous vehicle (AV) is navigating. A method includes receiving environment data from at least one sensor in an AV control system mounted to the AV, the environment data including online tracking data that identifies at least one object within the environment data that is recorded at drive time; annotating the at least one object from the environment data that are incorrectly identified by the AV control system; executing an offline tracking engine to generate offline tracking data that tracks the objects over time in the environment data; and identifying safety gaps between the online tracking data and the offline tracking data.

Inference and explanation from LLM:

YES. The invention described involves the use of sensors and tracking engines to identify and track objects in the environment of an autonomous vehicle (AV). This process inherently requires some form of perception and understanding of the environment, which are key aspects of artificial intelligence. The system uses both online and offline tracking mechanisms to ensure accurate identification of objects, indicating a level of cognitive processing and learning. The annotation step also suggests an improvement loop where the AI learns from its mistakes, enhancing its perceptual abilities over time. Therefore, this invention is indeed related to artificial intelligence with perceptual capabilities.

Title: IMAGE PROCESSING APPARATUS, IMAGE PROCESSING METHOD, AND STORAGE MEDIUM

Patent Application Number: 20220172470

Patent Applicant: Canon Inc.

Abstract: An image processing apparatus includes: a first designation unit configured to designate a first image group; a second designation unit configured to designate a second image group; an analysis unit configured to analyze each image included in the first image group and the second image group; a determination unit configured to determine a reference, based on an analysis result of the second image group; and a selection unit configured to select an image from the first image group, based on the reference and an analysis result of the first image group.

Inference and explanation from LLM:

NO. The described invention is related to image processing, which can be a component of artificial intelligence (AI) systems, but it does not necessarily imply perceptual ability as typically understood in AI contexts. The invention outlines a process for analyzing groups of images and selecting an image based on certain criteria, which could be part of a broader AI system. However, perceptual ability usually refers to more advanced cognitive processes such as recognizing objects, understanding scenes, or interpreting context, which go beyond mere analysis and selection based on predefined rules or references. This system seems to rely on pre-programmed functions rather than learning from experience or developing its own understanding of visual content, which would be necessary for true perceptual abilities in AI.

MOTOR CONTROL

Query: *Is the following invention directly related to artificial intelligence that has motor skills? Respond with YES or NO and give an explanation.*

Title: EMERGENCY STOPPING FOR AUTONOMOUS COMMERCIAL VEHICLES

Patent Application Number: 20180052463

Patent Applicant: Omnitracs LLC

Abstract: The present disclosure generally relates to autonomous commercial vehicles. In one aspect, the disclosure provides a method for controlling a commercial highway vehicle. The method includes detecting a failure of a first component based on a first signal from a first sensor. The method also includes classifying, by an automated driving system on the vehicle, a severity of the component failure. The method further includes determining to stop the vehicle if the severity exceeds a threshold severity level. The method also includes determining an emergency stopping distance based on the severity and a current momentum of the vehicle. The method further includes determining a stopping location within the emergency stopping distance. The method also includes stopping the vehicle at the stopping location. The present disclosure also provides an autonomous commercial vehicle and an emergency control system for performing the method.

Inference and explanation from LLM:

YES. The invention described involves an autonomous commercial vehicle that can detect and classify failures in its components, determine appropriate actions based on the severity of those failures, and perform actions such as stopping the vehicle safely. This process involves the use of sensors to gather data (like the status of components), automated systems to analyze this data and make decisions (such as whether a failure is severe enough to warrant stopping), and then executing those decisions (like stopping the vehicle). These functionalities require advanced computational capabilities and decision-making processes, which are hallmarks of artificial intelligence. Additionally, the act of stopping the vehicle at a determined location involves motor skills or precise control over the vehicle's movement, demonstrating AI integration with physical action.

Title: METHOD AND SYSTEM FOR REAL-TIME AND OFFLINE DE-IDENTIFICATION OF FACIAL REGIONS FROM REGULAR AND OCCLUDED COLOR VIDEO STREAMS OBTAINED DURING DIAGNOSTIC MEDICAL PROCEDURES

Patent Application Number: 20210209734

Patent Applicant: GE Precision Healthcare LLC

Abstract: Systems and techniques that facilitate real-time and/or offline de-identification of facial regions from regular and/or occluded color video streams obtained during diagnostic medical procedures are provided. A detection component can generate a bounding box substantially around a person in a frame of a video stream, can generate a heatmap showing key points or anatomical masks of the person based on the bounding box, and can localize a face or facial region of the person based on the key points or anatomical masks. An anonymization component can anonymize pixels in the frame that correspond to the face or facial region. A tracking component can track the face or facial region in a subsequent frame based on a structural similarity index between the frame and the subsequent frame being above a threshold. If the structural similarity index between the frame and the subsequent frame is above the threshold, the tracking component can track the face or facial region in the subsequent frame without having the detection component generate a bounding box or a heatmap in the subsequent frame, and the anonymization component can anonymize pixels in the subsequent frame corresponding to the tracked face or facial region.

Inference and explanation from LLM:

NO. The described invention is not directly related to artificial intelligence with motor skills. The system and techniques mentioned focus on de-identifying facial regions from video streams, which involves computer vision and image processing tasks such as object detection (face localization), heatmap generation for key points, and pixel anonymization. While these tasks require sophisticated AI algorithms, they do not involve motor skills, which typically refer to the ability to control physical movements or manipulate objects in the environment. This invention is more about privacy protection and data security in medical imaging rather than interacting with or controlling physical devices or systems.

LANGUAGE

Query: *Is the following invention directly related to artificial intelligence that has language ability? Respond with YES or NO and give an explanation.*

Title: DISPLAY-BASED CONTEXTUAL NATURAL LANGUAGE PROCESSING

Patent Application Number: 20220246139

Patent Applicant: Amazon Technologies Inc.

Abstract: Multi-modal natural language processing systems are provided. Some systems are context-aware systems that use multi-modal data to improve the accuracy of natural language understanding as it is applied to spoken language input. Machine learning architectures are provided that jointly model spoken language input (utterances) and information displayed on a visual display (on-screen information). Such machine learning architectures can improve upon,

and solve problems inherent in, existing spoken language understanding systems that operate in multi-modal contexts.

Inference and explanation from LLM:

YES. The described invention is directly related to artificial intelligence with language ability because it involves multi-modal natural language processing systems. These systems use both spoken language input and visual information to enhance natural language understanding, which falls under AI technologies that process and understand human language. Additionally, the use of machine learning architectures to model these inputs indicates an advanced form of AI capable of learning from various data types to improve its performance in understanding and processing natural language in multi-modal contexts.

Title: VIDEO ANALYSIS USING A DEEP FUSION REASONING ENGINE (DFRE)

Patent Application Number: 20220351521

Patent Applicant: Cisco Technology Inc.

Abstract: In one embodiment, a video analysis service receives video data captured by one or more cameras at a particular location. The service applies a neural network-based model to portions of the video data, to identify objects within the video data. The service maps outputs of the neural network-based model to symbols using a conceptual space. The outputs of the model comprise the identified objects. The service applies a symbolic reasoning engine to the symbols, to generate an alert. The service sends the alert to a user interface in conjunction with the video data.

Inference and explanation from LLM:

NO. While the described invention involves elements of artificial intelligence, such as neural networks for object identification, it does not explicitly involve language abilities. The invention focuses on processing video data to identify objects and then uses a symbolic reasoning engine to generate alerts based on those identifications. Although mapping outputs to symbols could be seen as a form of representation that might involve some level of abstraction or conceptual understanding, it does not necessarily imply language ability. Language ability typically involves more complex interactions such as understanding, generating, and interpreting natural language text or speech, which is not evident in this description.

DECISION-MAKING:

Query: *Is the following invention directly related to artificial intelligence that has decision-making ability? Respond with YES or NO and give an explanation.*

Title: SYSTEM AND METHOD FOR TRACKING EGRESS TIMES FROM A PARKING FACILITY AND PROVIDING ACTION RECOMMENDATIONS

Patent Application Number: 20230141416

Patent Applicant: Honeywell International Inc.

Abstract: A method includes receiving vehicle data for a plurality of vehicles in a parking facility over time, the vehicle data including a time t1 indicating when each of the vehicles leaves its corresponding parking spot in the parking facility, and a time t2 indicating when each of the plurality of vehicles reaches an exit of the parking facility. An egress time is determined for each of the plurality of vehicles based at least in part on the corresponding times t1 and t2. A timestamp is assigned to each of the egress times, resulting in timestamped egress times. A current egress time of the parking facility is estimated based at least in part on the timestamped egress times. An action recommendation is based at least in part on the current egress time of the parking facility and is outputted for subsequent viewing by the driver of the particular vehicle.

Inference and explanation from LLM:

YES. This invention involves collecting and processing data from vehicles within a parking facility to estimate the egress (exit) time of the facility. The system uses this data to make recommendations to drivers, which implies it has some form of decision-making capability. Specifically, the system analyzes historical data (times t1 and t2) to predict current conditions (current egress time) and then uses these predictions to suggest actions to drivers. This process involves pattern recognition, prediction, and decision-making, all of which are elements of artificial intelligence.

Title: METHOD AND SYSTEM FOR NAVIGATING AT A CLIENT DEVICE SELECTED FEATURES ON A NON-DYNAMIC IMAGE PAGE FROM AN ELASTIC VOICE CLOUD SERVER IN COMMUNICATION WITH A THIRD-PARTY SEARCH SERVICE

Patent Application Number: 20210099768

Patent Applicant: Dish Network LLC

Abstract: Systems, and methods are described for navigating non-dynamic pages at a client device served up from voice search requests by selecting a feature on the image page by a user request for navigating the feature viewed on the image page wherein the feature selected has an appearance of a dynamic feature when served up on the display wherein the user request selects a location on the image page; identifying the feature by the voice cloud server by location data of a selected location on the page; executing a feature request on a corresponding page which contains dynamic script at the voice cloud server to receive a linked page from a third-party search service; and removing dynamic scripts, at the voice cloud search server, of the linked page for serving up as an image page of the linked page at the client device.

Inference and explanation from LLM:

NO. The described invention involves navigating through web pages using voice commands and ensuring that the resulting pages are displayed without dynamic content (such as JavaScript) for better performance or accessibility. While it uses AI in the form of voice recognition to interpret user requests, the core function is not about AI making decisions but rather about processing user commands and serving simplified versions of web pages based on those commands. The decision-making aspect here is relatively straightforward and does not involve complex reasoning or learning capabilities typical of advanced AI systems.

LEARNING

Query: *Is the following invention directly related to artificial intelligence that has learning ability? Respond with YES or NO and give an explanation.*

Title: SYSTEMS AND METHODS FOR UTILIZING TIME SERIES AND NEURAL NETWORK MODELS TO DEPLOY AUTONOMOUS VEHICLES FOR 5G NETWORK COVERAGE GAPS

Patent Application Number: 20210297878

Patent Applicant: Verizon Patent and Licensing Inc.

Abstract: A device may receive historical usage data associated with a portion of a wireless network, and may train a model with the historical usage data to generate a trained model. The device may process data identifying a future time period, with the trained model, to forecast a traffic demand during the future time period, and may compare the traffic demand and a traffic capacity of the portion of the wireless network to determine whether the traffic demand is within a threshold of or exceeds the traffic capacity. The device may identify an autonomous vehicle to deploy for the portion of the wireless network when the traffic demand is within the threshold of or exceeds the traffic capacity, and may cause the autonomous vehicle to be deployed for the portion of the wireless network, wherein the autonomous vehicle is to provide a temporary wireless network.

Inference and explanation from LLM:

YES. The described invention involves the use of a model that is trained on historical usage data from a wireless network, which indicates a form of machine learning where the system can analyze past patterns to make predictions about future traffic demands. This process of training a model on data and using it to make forecasts is a fundamental aspect of artificial intelligence (AI) with learning abilities. Furthermore, the system uses these predictions to make decisions about deploying resources (in this case, an autonomous vehicle providing a temporary wireless network), which showcases an AI's ability to not only learn but also to take actions based on learned insights. Therefore, the invention is directly related to AI that has learning capabilities.

Title: GEORADIO

Patent Application Number: 20220067080

Patent Applicant: Gotham Studios Inc.

Abstract: Systems, methods, and computer-program products are described for determining current location information which includes determining whether an object is moving by comparing the current location information with previous location information. Future location information is calculated and a point of interest is filtered using the future location information. A media asset is then matched to the filtered point of interest and a current score is determined for the matched media asset. Determining a current score includes determining a window in which the matched media asset is geographically relevant to the future location information, and updating the matched media asset score when the future location information changes. An ordered playlist

is dynamically generated according to the future location information by repeatedly prioritizing and sequencing matched media assets according to the current score for each matched media asset, and matched media assets are played or displayed in the ordered playlist.

Inference and explanation from LLM:

NO. While the system described involves sophisticated algorithms for predicting future locations and dynamically adjusting content based on predicted user movement, it does not explicitly describe a learning process where the system improves its performance over time through experience. The described system relies on predefined rules and calculations rather than a self-improving learning mechanism typical of AI systems with learning abilities. It uses current and future location information to filter points of interest and match media assets, but this is more about predictive analytics and dynamic content adaptation than learning from interactions or data feedback to improve its performance iteratively.

ENGAGEMENT

Query: *Is the following invention directly related to artificial intelligence that has the ability to engage with users? Respond with YES or NO and give an explanation.*

Title: INITIALIZING A CONVERSATION WITH AN AUTOMATED AGENT VIA SELECTABLE GRAPHICAL ELEMENT

Patent Application Number: 20230336505

Patent Applicant: Google LLC

Abstract: Methods, apparatus, systems, and computer-readable media are provided for invoking an agent module in an automated assistant application in response to user selection of a selectable element presented at a graphical user interface rendered by a non-automated assistant application. The invoked agent module can be associated with other content rendered in the non-automated assistant graphical user interface, and can optionally be invoked with values that are based on user interactions via the non-automated assistant application. Responsive content can be received from the agent module in response to the invocation, and corresponding content provided by the automated assistant application via an automated assistant interface. In these and other manners, selection of the selectable element causes transition from a non-conversational interface, to a conversational automated assistant interface—where an agent (relevant to content in the non-conversational interface) is invoked in the automated assistant interface.

Inference and explanation from LLM:

YES. The described invention is indeed directly related to artificial intelligence as it involves engaging with users through an automated assistant application. This system allows for the transition from a non-conversational interface to a conversational one where an AI agent can be invoked based on user interaction within another application. The agent module's ability to provide relevant responses based on user interactions and the context of the interface indicates the use of AI technologies such as natural language processing (NLP), machine learning, and possibly more advanced AI techniques for understanding context and generating appropriate responses. Therefore,

this invention is clearly designed to facilitate user engagement through intelligent, interactive means, which is a hallmark of AI-based inventions.

Title: DYNAMICALLY RECONFIGURABLE INSURANCE PRODUCT

Patent Application Number: 20210258756

Patent Applicant: Accenture Global Solutions Ltd.

Abstract: A location-based risk alerting system is disclosed. When information regarding a risk assessment situation is received, the location and time attributes of the risk assessment situation are extracted, a risk score is calculated and a risk alert is generated to include the attributes and the risk score upon validating the received information for risk. The risk alert is transmitted to client devices of subscribers where a user-specific risk score is calculated at each client device based on a comparison of the distance of the client device from the location attribute and the time at which the risk alert is received with the time attribute. The risk alert is displayed at the client device based on the user-specific risk score and information regarding the user is shared with trusted authorities upon receiving user consent.

Inference and explanation from LLM:

NO. While this invention involves sophisticated processing and decision-making, it does not necessarily imply direct engagement with users in the sense typical of AI systems designed for interaction, such as chatbots or virtual assistants. Instead, it primarily focuses on automated processes for generating and distributing alerts based on predefined criteria (location, time, risk assessment). Although there is some user interaction implied in the context of receiving alerts and potentially sharing information with authorities, the core functionality described does not inherently require engaging with users through natural language processing, learning from interactions, or other AI-driven conversational techniques.

CREATIVITY

Query: *Is the following invention directly related to artificial intelligence that can generate content? Respond with YES or NO and give an explanation.*

Title: BODY FITTED ACCESSORY WITH PHYSICS SIMULATION

Patent Application Number: 20230074826

Patent Applicant: Snap Inc.

Abstract: Methods and systems are disclosed for performing operations comprising: receiving video that includes a depiction of a real-world object; generating a three-dimensional (3D) body mesh associated with the real-world object that tracks movement of the real-world object across frames of the video; obtaining an external mesh associated with an augmented reality (AR) element; determining that a first portion of the external mesh is associated with movement information of the 3D body mesh; determining that a second portion of the external mesh is associated with movement information of an external force model; deforming the first and second portions of the

external mesh separately based on movement information of the 3D body mesh and movement information of the external force model; and modifying the video to include a display of the AR element based on the deformed first and second portions of the external mesh.

Inference and explanation from LLM:

YES. This invention involves the use of AI in generating and manipulating content, specifically augmented reality (AR) elements in videos. It uses machine learning techniques to track movements of real-world objects within a video and apply these movements to AR elements. The process of generating a 3D body mesh, determining how different parts of an AR element should move based on the movements of the real-world object and external forces, and then deforming and displaying these elements requires sophisticated algorithms that are typically powered by AI technologies. This demonstrates direct involvement with AI-generated content as it involves creating and integrating synthetic elements into real-world footage dynamically.

Title: AUTOMATED IMAGE ANALYSIS USING ARTIFICIAL INTELLIGENCE TECHNIQUES

Patent Application Number: 20230410476

Patent Applicant: Dell Products LP

Abstract: Methods, apparatus, and processor-readable storage media for automated image analysis using artificial intelligence techniques are provided herein. An example computer-implemented method includes obtaining image data associated with at least one event; characterizing a plurality of attributes of the image data by processing at least a portion of the image data using one or more artificial intelligence techniques and processing metadata associated with at least a portion of the image data using one or more rule-based techniques; generating a determination as to whether one or more portions of the image data have been modified based at least in part on the plurality of characterized attributes of the image data; and performing one or more automated actions, based at least in part on the determination, in connection with at least one user request related to the image data associated with the at least one event.

Inference and explanation from LLM:

NO. The described invention is related to automated image analysis using artificial intelligence (AI) techniques but does not specifically involve AI content generation. The focus is on analyzing images, detecting modifications within them, and taking automated actions based on this analysis. While it uses AI techniques for processing image data and metadata, the primary function is not about generating new content such as text, images, or other forms of media from scratch, which would be characteristic of AI content generation systems.

Table IA.3 Individual Workers' Skills

This table shows examples of skill sets held by individual workers undergoing job transitions within a firm or between a firm and its external labor market. Data on job transitions are drawn from Revelio Labs' inflow and outflow transitions datasets. Data on skill sets are compiled from Revelio Labs' individual user skills dataset. Panel A shows examples of skill sets held by newly-hired workers, while Panel B shows examples of skill sets held by workers departing a job.

Panel A. Examples of skills held by newly-hired workers					
Employee ID	Firm	Year	SOC 3-Digit Occupation	Within-Firm Transition?	Employee's Skill Set
49286590	NATIONAL INSTRUMENTS CORP	2015	Engineers (172)	Yes	analog, application-specific integrated circuits (ASIC), automation, C, C++, data acquisition, debugging, electrical engineering, electronics, embedded software, embedded systems, engineering, field-programmable gate arrays (FPGA), integrated circuits (IC), LabVIEW, manufacturing, MATLAB, PCB design, product development, product management, project management, semiconductors, soc, software tests, system design, test automation, test engineering, test equipment, testing
49279186	JPMORGAN CHASE & CO	2018	Other Educational Instruction and Library Occupations (259)	No	agile methodologies, C, C++, cascading style sheets (CSS), git, HTML, HTML5, Java, JavaScript, jQuery, Linux, MySQL, PHP, programming, Python, software development, structured query language, web development, XML
49267198	BOEING CO	2021	Media and Communication Workers (273)	No	altera, altera quartus, analog, application-specific integrated circuits (ASIC), C, C++, debugging, design specifications, digital signal processors, electronics, embedded software, embedded systems, field-programmable gate arrays (FPGA), integrated circuits (IC), logic design, MATLAB, microcontrollers, mixed signal, PCB design, PCIE, PERL, radio frequency (RF), requirements analysis, RTL design, SATA, semiconductors, serdes, simulations, SOC, statistical analysis software, systemverilog, testing, verilog, VHDL, XILINX, XILINX ISE

(Continued)

Table IA.3, Continued

Panel B. Examples of skills held by departing workers					
Employee ID	Firm	Year	SOC 3-Digit Occupation	Within-Firm Transition?	Employee's Skill Set
49282976	VMWARE INC-CL A	2010	Computer Occupations (151)	Yes	Cisco Technologies, cloud computing, data center, disaster recovery, information technology infrastructure library, integration, IT service management, Linux, network security, networking, requirements analysis, servers, software development, storage area networks, structured query language, TCP/IP, UNIX, virtualization, VMware, Windows Server
49250337	STARBUCKS CORP	2016	Supervisors of Food Preparation and Serving Workers (351)	No	catering, cooking, culinary skills, customer service, event planning, food, food & beverage, food safety, food service, hospitality, hospitality management, inventory management, leadership, management, menu development, Microsoft Excel, Microsoft Office, Microsoft Word, PowerPoint, process improvement, project management, research, restaurant management, restaurants, sales, team building, time management, training
49255333	INTEL CORPORATION	2021	Computer Occupations (151)	No	active server pages, agile methodologies, C, C++, databases, debugging, embedded software, embedded systems, hardware, integration, Java, JavaScript, Linux, management, Microsoft Office, MySQL, Perl, Python, software development, software engineering, structured query language, test automation, testing, Unix, web applications, Windows, wireless, XML

IA.4 Skill-Based AI Exposure and Other Alternatives

In this study, we measure occupational exposure to different types of AI using occupational skill requirements from the O*NET database, following a large literature that emphasizes the role of technology in augmenting or substituting for workers' skills and abilities (Acemoglu and Autor, 2011; Autor, Levy, and Murnane, 2003; Felten, Raj, and Seamans, 2019, 2021). An alternative approach in the literature measures technological exposure using occupational task descriptions from the O*NET database (e.g., Frey and Osborne, 2017; Webb, 2020; Eloundou et al., 2024; Eisfeldt et al., 2025; Hampole et al., 2026). Both O*NET datasets are at the occupational level but differ in what they capture. Specifically, skill requirements are 35 standardized capability descriptors rated uniformly across all occupations, whereas task descriptions are work activities highly specific to each occupation. Motivated by this distinction, we examine the feasibility of using task requirements to identify occupational exposure to different AI technologies. We provide both conceptual and empirical evidence showing that occupational skill requirements constitute a more appropriate unit of analysis for our setting, as AI systems primarily complement or substitute for underlying human capabilities that span multiple tasks rather than individual tasks themselves.

Tasks and skills are fundamentally different concepts in labor economics. Acemoglu and Autor (2011) define tasks as discrete units of work activity that produce output, characterizing what workers do on the job, and define skills as workers' endowments of capabilities for performing those tasks. Task descriptions such as processing customer payments, diagnosing medical conditions, or writing performance evaluations are therefore activity-level constructs that vary with the specific demands of a job. Skills such as inductive reasoning, social perceptiveness, or creativity are deeper human capabilities that workers possess and deploy across many different tasks and work contexts (Autor, 2013).

The existing task-based exposure literature addresses a different research question from our study. Frey and Osborne (2017) estimate the probability that each occupation will be automated, producing occupation-level computerization risk exposure. Webb (2020) constructs exposure measures by detecting textual overlap between occupation-specific task descriptions and narratives disclosed in patent filings. Eloundou et al. (2024) measure

occupation-level exposure to large language models by assessing the fraction of each occupation's tasks for which GPT can reduce the time required for completion. Eisfeldt et al. (2025) construct firm-level Generative AI exposure by applying a large language model to O*NET task descriptions and aggregating to the firm level via occupational workforce composition, further distinguishing core from supplemental task exposures to identify substitution versus complementarity effects. Each of these studies treats AI as a single or broadly unified technology and poses a feasibility question of whether that technology can perform or assist with a given task.

The task-based exposure in the literature is well-suited for quantifying occupational exposure to a broadly unified technology. However, since task descriptions characterize work at the output level without encoding which functional capabilities produce those outputs, mapping them to functionally distinct AI types requires an additional inference that introduces ambiguity. Specifically, task descriptions bundle multiple heterogeneous functional capabilities into work activities. For instance, the O*NET task *“Use gestures to shape the music being played, communicating desired tempo, phrasing, tone, color, pitch, volume, and other performance aspects”* (Entertainers and Performers, Sports and Related Workers, SOC 272) simultaneously requires Motor Control (conducting gestures), Perception (real-time auditory monitoring of ensemble sound), Language (expressive communication of interpretive intent), Engagement (directing and coordinating performers), Decision-making (selecting desired values across multiple sonic dimensions), Learning (adaptively adjusting based on the ensemble's response), and Creativity (artistic shaping of musical interpretation), making it infeasible to attribute task exposure to a specific AI functional type. Since most tasks involve multiple cognitive dimensions, mapping tasks to our functional AI typology may yield an indiscriminate result.

In contrast, occupational skill requirements from O*NET are defined at the level of individual capabilities and therefore correspond directly to the functional distinctions in our AI typology. Skills are defined as workers' endowments of capabilities from which task performance is derived, making them the more fundamental unit of analysis (Acemoglu and Autor, 2011). Our AI typology classifies AI systems by the human capabilities they replicate or complement, and skill requirements are themselves capability descriptors. For instance, *“Critical Thinking”* is an O*NET skill that naturally maps to Learning AI only.

Therefore, grounding exposure measurement in skills directly captures the relatedness between an occupation's capability demands and the functional capabilities of each AI type.

To illustrate this issue, we conduct a mapping experiment for three large occupations whose skill-based AI exposure is particularly narrow: Retail Sales Workers (SOC 412, exposed to two AI types), Home Health and Nursing Aides (SOC 311, exposed to one AI type), and Motor Vehicle Operators (SOC 533, exposed to one AI type). These occupation groups collectively employ millions of workers. We identify the five most important O*NET tasks for each occupation (importance score ≥ 3) and map each task to the AI functional categories plausibly involved in performing it. The results are reported in Table IA.4.1. In Panel A, the top five tasks for Retail Sales Workers (i.e., greeting customers, processing payments, and recommending merchandise) collectively map to six out of seven AI categories. Similarly, Panel B shows that the top five tasks for Home Health and Nursing Aides (e.g., repositioning patients, answering call signals, feeding patients, and recording intake measurements) map to six of seven categories. As reported in Panel C, Motor Vehicle Operators, whose top tasks include traffic compliance and vehicle inspection, map to five of seven categories.

Compared with task-based mapping, our skill-based mapping reported in Table IA.4.2 reveals more information for cross-occupation variation on AI exposure. Our skill-based approach requires that at least four important skills map to an AI category before an occupation is classified as exposed. Under this criterion, the same three occupations exhibit sharply differentiated exposures to AI. In Panels A-C, Retail Sales Workers are exposed to only two AI types (Perception and Motor Control), Home Health and Nursing Aides to only one (Motor Control), and Motor Vehicle Operators to only one (Learning). This difference reflects meaningful diversity in cognitive demands. For instance, a retail salesperson's core competencies center on interpersonal perception and physical product handling; a nursing aide's on motor-intensive patient care; and a truck driver's on operational learning. Critically, this differentiation maps directly onto the distinction between displacing and augmenting AI that is central to our analysis. Retail Sales Workers and Home Health Aides are exposed exclusively to displacing AI types (Motor Control and Perception), while Motor Vehicle Operators are exposed exclusively to an augmenting AI type (Learning). Despite being similarly large blue-collar occupations, our skill-based approach reveals that

they face fundamentally different AI exposure with opposite predicted employment consequences. Table IA.4.3 summarizes these task-based mapping results alongside the corresponding skill-based exposures. The evidence in this section suggests that measuring AI exposure at the skill level allows us to better decompose occupational heterogeneity, thus improving the identification of the AI-displacing and AI-augmenting occupations.⁵

⁵ Following the spirit of recent studies (Brynjolfsson, Mitchell, and Rock, 2018; Eloundou et al., 2024), we also evaluate the suitability of using O*NET’s Generalized Work Activities (GWAs), a standardized set of 41 cross-occupational behaviors rated uniformly across all occupations, as a task-level benchmark for AI exposure. Applying identical threshold rules (at least four elements with importance ≥ 3 mapping to an AI type) across all 95 three-digit SOC occupations, the GWA-based approach classifies 62 of 95 occupations as exposed to six or seven AI types, with no occupation at zero. Our skill-based approach yields a far more dispersed distribution: 19 occupations at zero, 25 at one to three, 41 at four to five, and only 10 at six types. The GWA approach further assigns near-universal exposure to most AI types (e.g., Language, Decision-making), whereas the skill-based approach produces substantially more differentiated coverage across types, providing variation that is essential for identifying the heterogeneous labor-market effects of distinct AI functional types.

Table IA.4.1 Task-Based AI Mapping for Three Large Occupations

This table reports the job exposure results for three large occupations using task descriptions as mapping inputs. The top five O*NET task statements (rated by importance) for each occupation are manually mapped to seven AI functional types. ‘X’ indicates the task plausibly requires that AI capability.

Panel A. SOC 412 Retail Sales Workers

Task	Importance score	Perception	Motor Control	Language	Decision-making	Learning	Engagement	Creativity
Greet customers and ascertain what each customer wants or needs	4.84	X		X		X	X	
Receive payment by cash, check, credit cards, vouchers, or automatic debits	4.76	X	X		X	X		
Greet customers entering establishments	4.72			X			X	
Recommend, select, and help locate merchandise based on customer needs	4.68	X		X	X	X		
Issue receipts, refunds, credits, or change due to customers	4.68	X	X		X			

Panel B. SOC 311 Home Health and Nursing Aides

Task	Importance score	Perception	Motor Control	Language	Decision- making	Learning	Engagement	Creativity
Turn or reposition bedridden patients	4.76	X	X				X	
Answer patient call signals, signal lights, bells, or intercom systems	4.68	X	X	X			X	
Feed patients or assist patients to eat or drink	4.64	X	X				X	
Measure and record food and liquid intake or urinary and fecal output	4.60	X		X	X	X		
Provide physical support to assist patients to perform daily living activities	4.52	X	X		X		X	

Panel C. SOC 533 Motor Vehicle Operators

Task	Importance score	Perception	Motor Control	Language	Decision-making	Learning	Engagement	Creativity
Comply with traffic regulations to operate vehicles in a safe and courteous manner	5.00	X	X		X	X		
Follow safety rules as students board and exit buses or cross streets near bus stops	4.93	X	X		X	X	X	
Check the condition of a vehicle's tires, brakes, windshield wipers, lights, oil, fuel, water, and safety equipment	4.93	X	X		X	X		
Test vehicle equipment, such as lights, brakes, horns, or windshield wipers, to ensure proper operation	4.84	X	X		X			
Obey traffic laws and follow established traffic and transportation procedures	4.83	X	X		X	X		

Table IA.4.2 Skill-Based AI Mapping and Detailed Skill-Level Results

This table reports O*NET skill requirements with importance ≥ 3 and the AI functional types to which each skill maps for three major occupations reported in Table IA.4.1. An occupation is classified as exposed to an AI type if four or more of its skills map to that type.

Panel A. SOC 412 Retail Sales Workers

Skill	Importance	AI Types Mapped
Active Listening	3.50	Perception, Motor Control, Engagement
Speaking	3.43	—
Service Orientation	3.37	Perception, Motor Control, Language, Learning
Social Perceptiveness	3.20	Perception, Motor Control, Language, Engagement
Reading Comprehension	3.15	Perception, Motor Control, Language,
Persuasion	3.10	Engagement
Critical Thinking	3.02	Learning

Panel B. SOC 311 Home Health and Nursing Aides

Skill	Importance	AI Types Mapped
Service Orientation	3.78	Perception, Motor Control, Language, Learning
Social Perceptiveness	3.65	Perception, Motor Control, Language, Engagement
Active Listening	3.52	Perception, Motor Control, Engagement
Monitoring	3.22	Motor Control, Language, Decision-Making, Learning
Speaking	3.22	—
Critical Thinking	3.18	Learning
Coordination	3.07	—

Panel C. SOC 533 Motor Vehicle Operators

Skill	Importance	AI Types Mapped
Operation and Control	3.30	Decision-Making, Learning, Engagement
Active Listening	3.25	Perception, Motor Control, Engagement
Speaking	3.17	—
Critical Thinking	3.15	Learning
Operations Monitoring	3.07	Decision-Making, Learning, Engagement
Monitoring	3.02	Motor Control, Language, Decision-Making, Learning
Time Management	3.00	—

Table IA.4.3 Comparison of Task-Based and Skill-Based AI Exposure

This table summarizes the number of AI types each occupation is classified as exposed to under the task-based approach (top five O*NET task statements) and the skill-based approach (O*NET skill requirements, threshold ≥ 4 skills with importance ≥ 3).

SOC	Occupation	Task-Based	Skill-Based
412	Retail Sales Workers	6/7	2/7
311	Home Health and Nursing Aides	6/7	1/7
533	Motor Vehicle Operators	5/7	1/7